

Rahul Reza Roky

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EXPERIENCE

Software Engineer (Backend) | Pathao

Dhaka, Bangladesh | Jan 2024 – Present

- Architected authentication system for **Pathao Offline Ride** with caching and multi-layer rate limiting, achieving **30ms P99 latency** serving **1M+ users** with **99.9% uptime**
- Engineered **Pathao Rent-a-Car** manual ride management backend with optimized PostgreSQL indexing and Redis caching, leveraging Go concurrency for **sub-50ms API responses**
- Built **n-gram fuzzy search** engine powering Pathao's primary custom address builder, serving **millions of users** with real-time location matching
- Hunted a production Go allocation leak with **pprof** — **85% heap churn reduction (16.3 GB → 2.5 GB)** in under 1 hour, live at **~100 RPS**, zero downtime
- Designed modular **GitLab CI pipeline** automating GitOps deployments for **20+ microservices** using job artifacts and conditional rules
- Established testing framework with testify and GoMock achieving **85%+ coverage**, reducing production incidents by **40%**

EDUCATION

B.Sc. in Computer Science and Engineering

Rajshahi University of Engineering & Technology (RUET)

2018 – 2023

CGPA: 3.23/4.0

PROJECTS

Enterprise URL Shortener | Go, PostgreSQL, Redis, Docker

github.com/RahulReza920/url-shortener

- Built production-grade URL shortener with **Clean Architecture**, implementing PASETO v4 authentication and distributed rate limiting (GCRA)
- Designed collision-free ID generation using bijective encoding with salt-based shuffling; achieved **< 10ms P95 latency**
- Implemented comprehensive testing suite with unit tests, integration tests, and E2E verification scripts via Makefile automation

Real-Time Chat Application | Rust, Axum, Tokio, WebSockets

github.com/RahulReza920/chat_app

- Developed concurrent chat server using Tokio async runtime and Axum framework with safe WebSocket handling
- Utilized broadcast channels for real-time message distribution across multiple connected clients with structured tracing

TECHNICAL SKILLS

- **Languages:** Go, C/C++, Rust
- **Backend:** gRPC, REST API, RabbitMQ, Google Pub/Sub, Redis
- **Databases:** PostgreSQL, MySQL
- **Infrastructure:** Docker, Kubernetes, GCP, Consul, GitLab CI/CD
- **AI/ML Expertise:** PyTorch, TensorFlow, NLP, Computer Vision, Deep Learning, Data Mining
- **Tools:** Git, Makefile, Linux/Bash, Jupyter Notebook, Google Colab, Anaconda

PUBLICATIONS

- **BHSGR-Net: A Light-Weight Convolutional Neural Architecture for Recognition of Bengali Hand Sign Gestures**
International Conference on Big Data, IoT and Machine Learning (BIM 2023) [\[Link\]](#)
- **Empowering the Deaf Community: A Vision-Based Method for Understanding Bangla Sign Language Numerals.** [Undergraduate Thesis]

ACHIEVEMENTS

- Kubernetes for the Absolute Beginners - Hands-on Certificate – KodeKloud [\[Certificate\]](#)
- Technical blog with **20,000+ monthly readers** and **130+ subscribers** on Medium [\[Profile\]](#)
- Top 5, RUET Intra University Programming Contest [\[Certificate\]](#)
- Machine Learning Specialization – Stanford University (Coursera) [\[Certificate\]](#)